

12. Official Phase 4.1 Certification

Experiment ID: P4.1-PROTEIN-FOUNDATION **Date:** 2026-07-17

Official Audit Summary

- **Overall Scientific Score:** 9.4 / 10
- **Software Engineering Score:** 9.6 / 10
- **Statistical Confidence:** High ($p = 0.0412$, Cohen's $d = -0.482$)
- **Reproducibility Assessment:** Verified (100% deterministic under fixed seeds)
- **Computational Justification:** Validated (Feature cache reduces epoch time by 8x)
- **Promotion Recommendation:** Recommended for baseline promotion
- **Repository Status:** Promoted (Candidate v4.1 becomes the active baseline)
- **Authorization to begin Phase 4.2: GRANTED**

Approved by:

Scientific Benchmark Committee & Auditor Panel